

SCI-NOI v0.1

SCI-NOI v0.1 r0.2 Shared Normative Core

Implementation-blind Stage B draft

SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.2

Not owner-accepted or frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

SCI-NOI v0.1 r0.2 Shared Normative Core

Document identity: SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.2

Status: implementation-blind Stage B draft; not owner-accepted or frozen.

The complete shared normative authority is the following six files in this fixed order, byte-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2:

1. normative/NOTATION.md
2. normative/DEFINITIONS.md
3. normative/EQUATIONS.md
4. normative/ASSUMPTIONS.md
5. normative/REQUIREMENTS.md
6. normative/PREDICTIONS.md

Binding file: NORMATIVE_MODULE_BINDING.json

Binding-file SHA-256: 5c6f954b457a546abcf74a1dc6dae190f2b22ea43c14edf34d2e4d2a8a704268.

Each module's exact digest appears in that binding file. A byte change to any module requires a new binding digest and a new document generation. No summary, rationale, ECS procedure, table, or PDF rendering supersedes the modules. The normative-core PDF appends their exact text in the stated order.

Authority boundary

The core is bound to SCI-NOI_AUTHOR_PACKET_MANIFEST v0.1/r0.18, SHA-256 b6f8e7252e7f61f4506899cb3e8e26cf939887bb48464852713f8ce81ac77ca0, and its controlling SCI-NOI_OWNER_DECISIONS v0.1/r0.18, SHA-256 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b. Authorized SCI-VAL successor records are process-only binding evidence and add no NOI scientific content.

Draft ceiling

The core is an implementation-blind proposal. It does not establish implementation conformity, validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, or production authorization. Explicit pending owner questions remain unavailable rather than receiving invented defaults.

Appendix A. Exact Bound Normative Modules

SCI-NOI v0.1 r0.2 Shared Normative Module: Notation

Module identity: SCI-NOI_NORMATIVE_NOTATION v0.1/r0.2

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2; not owner-accepted or frozen.

- Symbol | Exact meaning
- i | One exact retained PTC occurrence, with immutable observation, detector/channel, stable RTC sample, product/application generation, segment, array/network/group, ancestry, time, validity, and provenance identity.
- d(i) | Stable realized detector/channel identity owning occurrence i.
- d | One stable realized detector/channel within one exact observation. It is the approved ordinary coherence unit.

- $h(d)$ | Stable readout-network stratum containing detector d for one exact observation. Missing or ambiguous membership is unavailable.
- b | One admitted realization-member identity in one exact design generation.
- p | One exact frozen MAP output-row/pixel identity on its declared WCS, frame, support, response, and lifecycle domain.
- Z_i^{PTC} | Exact PTC transformed signal for occurrence i . The superscript and owner are mandatory.
- γ_i | Exact PTC-owner-selected MAP-facing coefficient for occurrence i , including family, generation, value, and QC state.
- G_{pi} | Exact frozen MAP projection/gridding contribution from occurrence i to output row p .
- a_{pi} | Exact frozen positive pre-normalization MAP contribution $G_{pi} \gamma_i$ on an admitted occurrence/row pair.
- C_p | Exact frozen set of admitted PTC occurrences contributing to MAP output row p .
- Q_p | Exact unsigned frozen MAP normalization $\sum_{i \in C_p} a_{pi}$. It is not a signal quantity and is never signed or relearned by NOI.
- β_d | Canonical r0.2 detector coefficient mass $\sum_p \sum_{i \in C_p, d(i)=d} a_{pi}$. It replaces the r0.1 spelling B_d to avoid collision with member counts.
- ϵ_{bd} | Canonical r0.2 sign assigned to detector d in member b , in $\{-1, +1\}$. It replaces the r0.1 spelling s_{bd} ; source quantities never use this symbol.
- $\text{mathcal{D}}$ | Complete finite assignment-design identity: target law, population, ordering, construction, counts, dependence, equivalence, rank, weights, keying, persistence, and generation. It replaces the ambiguous r0.1 bare S .
- $M_b(p)$ | NOI realization-map value for member b at row p , produced by the exact ordinary frozen MAP operator equation. It is not an ordinary MAP science product.
- $m_{\text{MAP}}(p)$ | Exact immutable normalized real-observation MAP signal, using the corresponding all-+1 operation and retaining ordinary MAP product identity. It replaces the r0.1 spelling q_{MAP} .
- $V_{\text{hat_cond}}(p)$ | Initial zero-centered design-weighted conditional detector-sign-randomization second moment on the exact common all-member domain. It replaces the r0.1 spelling $V_{\text{hat_cond}}$.
- $\rho_{\text{cond}}(p)$ | Proposed r0.2 successor spelling for the reciprocal $1/V_{\text{hat_cond}}(p)$. It is unavailable pending exact owner acceptance and successor profile/source binding. The owner-approved r0.18 legacy method identity remains immutable history.
- $\sigma_{\text{cond}}(p)$ | Canonical positive denominator $\sqrt{V_{\text{hat_cond}}(p)}$ on the exact finite-positive compatible domain.
- $\zeta_{\text{cond}}(p)$ | Canonical r0.2 standardized product $m_{\text{MAP}}(p)/\sigma_{\text{cond}}(p)$. It replaces the r0.1 spelling S_{cond} and has unit exactly 1.
- $N_{\text{requested}}$ | Requested member count, preregistered before candidate or member output is inspected.
- N_{resolved} | Member count in the successfully resolved design.
- $N_{\text{completed}}$ | Count of members that reached the producer's complete terminal state.
- N_{admitted} | Count admitted to one exact named UNC method after member and ensemble policy evaluation.
- $N_{\text{unique_sign}}$ | Count of byte-identical-distinct canonical signed assignment vectors.
- $N_{\text{unique_orbit}}$ | Count of complement orbits $\{\epsilon, -\epsilon\}$ on the identical ordered active domain.
- r_{sign} | Exact rank of the method-declared sign-design operator on the active coherence domain.
- r_{map} | Exact rank, or declared rank limit/unavailable state, after projection of the admitted design into the exact map-domain statistic or covariance representation.

- `I_attempt` | Execution-attempt/evidence identity. It may change during an idempotent rerun without changing scientific member identity.

Inactive r0.1 spellings appear only in the change map or exact immutable upstream/profile quotations. They are not second active identities.

SCI-NOI v0.1 r0.2 Shared Normative Module: Definitions

Module identity: SCI-NOI_NORMATIVE_DEFINITIONS v0.1/r0.2

Status: implementation-blind Stage B draft authority; content-bound by
SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2; not owner-accepted or frozen.

D-001. Separate operations

NOI-GEN generates one declared finite realization ensemble. NOI-UNC infers one declared empirical uncertainty quantity from one exact admitted ensemble. NOI-STD joins one immutable signal numerator to one authorized positive uncertainty scale. No operation automatically realizes the next.

D-002. Ordinary route

The selected ordinary route is NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1. NOI inserts one assignment modifier at the exact PTC-to-MAP numerical boundary. MAP applies it inside otherwise frozen accumulation. NOI owns assignment, design, realization, and ensemble identity; MAP owns conforming application of its frozen arithmetic.

D-003. Fixed and relearned methods

A fixed method holds every scientifically consequential adjacent state fixed at Theta0. A relearned method identifies every consequential learn/resolve stage and its member-specific Theta_b. Different fixed/relearned classes or replay graphs are different ensemble identities and cannot mix.

D-004. Coherence populations

The complete coherence population contains every stable realized detector/channel identity in the exact observation/array parent domain. The active design population is the subset with exact positive beta_d and exact unambiguous network membership. Zero-mass, zero-total-stratum, singleton-active- stratum, and ambiguous-membership states remain separately typed; their final base-v0.1 design dispositions are pending owner acceptance of the complete finite-design candidate.

D-005. Source-bearing conditional randomization ensemble

This owner-authorized Stage B term denotes an ensemble whose fixed parent may contain astronomical source signal, structured residuals, and source-model error. The design balances network-local frozen pre-normalization MAP coefficient mass. Under additional source and coverage assumptions this may reduce some detector-common source contribution in aggregate. It does not establish pixelwise source cancellation, a source-free null, calibrated null, or repeated physical-noise sampling.

D-006. Scientific replacement and execution retry

A scientific replacement changes an assignment, plan, parent, frozen operator, or scientifically substituted member. It requires a new member and every applicable new design/generation identity. An execution retry repeats identical deterministic work after an infrastructure interruption. It may retain member, assignment, parent, operator, and scientific generation only when a new I_attempt records the execution and no scientific state changes.

D-007. Initial UNC target

The initial target is the conditional detector-sign-randomization law for one exact fixed-state parent/operator/design. Complement symmetry makes its target- law mean zero. Its marginal second moment therefore equals its conditional- randomization marginal variance. This is neither repeated-observation physical-noise variance nor total MAP uncertainty.

D-008. Common all-member domain

The initial UNC domain contains exactly the map rows where every admitted member supplies a valid finite value under one identical member population. Survivor, pairwise, zero-filled, interpolated, and domain-extended substitutes are different methods and are not supplied by base v0.1.

D-009. Covariance and rank plurality

A pointwise second moment, retained ensemble, marginal variance, fixed projection, structured covariance, full covariance, reciprocal, precision, and consumer weight are different product roles. `r_sign` describes the sign design; `r_map` describes its exact projected information for a named map-domain use. Neither count nor one rank substitutes for the other.

D-010. Reciprocal disposition

ODQ-107 approved a finite-positive reciprocal in r0.18. r0.2 removes the misleading active phrase "inverse ... scale" and proposes the sole successor role `reciprocal_conditional_randomization_second_moment`. Because this changes the approved identity/profile spelling, the r0.2 successor remains unavailable pending the stable owner question and new immutable profile/source binding. Inverse variance, precision, and consumer weight remain unavailable.

D-011. Standardized product

The initial standardized product is `zeta_cond`, the exact immutable normalized real-observation MAP signal divided by `sqrt(Vhat_cond)` on their exact compatible finite-positive domain. Its unit is 1. It is not uncertainty, Gaussian/Student/pivotal/z/N-sigma significance, probability, detection, false-alarm rate, completeness, purity, or catalog authority.

D-012. Persistence modes

ODQ-109 authorizes three plan-selected modes: persisted ensemble, compact deterministic regeneration, and streaming sufficient statistics. A composite of more than one is a distinct plan. Each mode declares exact reconstruction, numerical regeneration, retained sufficiency, unavailable later questions, lifecycle, failure, and completion. No default or silent fallback exists.

D-013. Profiles and Registry

GEN owns completion truth. NOI owns each named-use admission policy. SCI-VAL binds and evaluates but authors neither. The four Stage A profile bytes are owner-approved and exactly registered. r0.2 notation or action-byte changes require an immutable profile/Registry successor before r0.2 evaluation; a profile name alone is never an eligibility decision.

SCI-NOI v0.1 r0.2 Shared Normative Module: Equations

Module identity: SCI-NOI_NORMATIVE_EQUATIONS v0.1/r0.2

Status: implementation-blind Stage B draft authority; content-bound by
SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2; not owner-accepted or frozen.

SCI-NOI-EQ-001 - frozen pre-normalization contribution

$$a_{pi} = G_{pi} \gamma_i.$$

The factors, occurrence membership, coefficient generation, and QC state are exact frozen parent facts.

SCI-NOI-EQ-002 - frozen unsigned normalization

$$Q_p = \sum_{i \in C_p} a_{pi}.$$

Q_p is evaluated only on the exact frozen MAP output-row domain where its parent gates are satisfied. It is not signed or relearned.

SCI-NOI-EQ-003 - detector coefficient mass

$$\beta_d = \sum_p \sum_{i \in C_p, d(i)=d} a_{pi}.$$

The sum uses the exact frozen admitted occurrence/row population. β_d replaces the ambiguous r0.1 symbol B_d .

SCI-NOI-EQ-004 - candidate network imbalance

$$\begin{aligned} \Delta_h(\epsilon_b) &= \text{abs}(\sum_{d \in D_h^{\text{active}}} \epsilon_{bd} \beta_d) \\ &\quad / \sum_{d \in D_h^{\text{active}}} \beta_d. \end{aligned}$$

This equation is an author-recommended ODQ-102D candidate, not accepted base science. It shall not be evaluated when the denominator is zero. Its tolerance, population, construction, and failure semantics remain unavailable pending owner acceptance of the complete design candidate.

SCI-NOI-EQ-005 - canonical ordinary realization member

$$\begin{aligned} M_b(p) &= [\sum_{i \in C_p} a_{pi} \epsilon_{b,d(i)} Z_i^{\text{PTC}}] / Q_p. \end{aligned}$$

The sign modifies only Z_i^{PTC} . G_{pi} , γ_i , C_p , Q_p , WCS, projection, support, coverage_cut, response handling, and every MAP-local gate remain frozen. The denominator is unsigned and is neither randomized nor relearned. The sign is applied exactly once to each admitted occurrence. Inline and materialized application are equivalent only under this equation and identical parent/membership identity.

SCI-NOI-EQ-006 - corresponding real-observation MAP signal

$$m_{\text{MAP}}(p) = [\sum_{i \in C_p} a_{pi} Z_i^{\text{PTC}}] / Q_p.$$

This is the corresponding all-+1 operation and retains its own immutable ordinary MAP product identity. M_b is an NOI realization, not ordinary MAP science.

SCI-NOI-EQ-007 - fixed-state signal/response operator relation

$$R_b^{\text{fixed}} = A_{\text{MAP},Pi} D_{\epsilon_b} H_{\text{PTC}}^{\text{fixed}}.$$

H_{PTC}^{fixed} maps exact admitted parent occurrences into the fixed PTC output domain; D_{ϵ_b} applies the same detector assignment exactly once; $A_{MAP, \Pi}$ applies the exact frozen MAP projection, membership, normalization, support, response-reference, and publication operator to the declared output domain Π . A response companion, when required and available, receives the identical occurrence membership and identical modifier exactly once under its owner-declared response operator. Missing response is typed unavailable and shall not create a hidden subset of signal membership.

SCI-NOI-EQ-008 - complement-symmetric target mean

$$\begin{aligned} E_D[\epsilon_{bd}] &= 0, \\ E_D[M_b(p) \mid \text{fixed parent and operator}] &= 0. \end{aligned}$$

The second equality follows from the linear fixed-state member equation and the approved complement-symmetric target law. It does not remove source or structured residual terms from individual members, second moments, or tails.

SCI-NOI-EQ-009 - initial conditional second moment

$$\begin{aligned} \text{Vhat_cond}(p) &= \sum_{b \in A_{UNC}} \omega_b M_b(p)^2, \\ \sum_b \omega_b &= 1, \\ \omega_b &\geq 0. \end{aligned}$$

The finite design supplies the exact weights. The center is the known target zero; the finite ensemble mean is not subtracted and no B-1 rule applies. The equation is evaluated only on one exact common all-member domain.

SCI-NOI-EQ-010 - proposed reciprocal successor

$$\text{rho_cond}(p) = 1 / \text{Vhat_cond}(p).$$

This r0.2 successor spelling is unavailable pending owner acceptance and profile/source rebinding. If accepted, its domain is only the exact finite strictly positive parent domain. It is not inverse variance, precision, validity, support, exposure, a consumer weight, or a PTC/MAP coefficient.

SCI-NOI-EQ-011 - canonical conditional scale

$$\text{sigma_cond}(p) = \sqrt{\text{Vhat_cond}(p)}.$$

The scale exists only on the exact finite-positive compatible parent domain and has the signal unit of M_b and m_{MAP} .

SCI-NOI-EQ-012 - canonical standardized MAP signal

$$\text{zeta_cond}(p) = m_{MAP}(p) / \text{sigma_cond}(p).$$

The output exists only on the exact compatible finite-positive intersection, has unit exactly 1, and carries the claim in SCI-NOI-REQ-031 only.

SCI-NOI v0.1 r0.2 Shared Normative Module: Assumptions

Module identity: SCI-NOI_NORMATIVE_ASSUMPTIONS v0.1/r0.2

Status: implementation-blind Stage B draft authority; content-bound by
SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2; not owner-accepted or frozen.

SCI-NOI-ASM-001 - exact authority

Scientific authority is restricted to the 17 objects in SCI-NOI_AUTHOR_PACKET_MANIFEST v0.1/r0.18, SHA-256 b6f8e7252e7f61f4506899cb3e8e26cf939887bb48464852713f8ce81ac77ca0. The controlling owner artifact is SCI-NOI_OWNER_DECISIONS v0.1/r0.18, SHA-256 272ac939b8a7109a123073b1a39fcd7ac4129c603683ee81257b94ab2f55a0b. Process-only SCI-VAL successor records may verify profile/source binding but author no NOI science.

SCI-NOI-ASM-002 - frozen numerical parent gates

The selected ordinary route remains numerically unavailable until its exact PTC MAP-facing coefficient family/value/QC, numerical coverage_cut state/value/failure policy, and every exact parent/admission fact are realized. No equation in this draft creates those parents or infers a default.

SCI-NOI-ASM-003 - fixed operator

For the ordinary method, the exact admitted occurrence population, coefficients, projection, normalization, WCS, support, boundary, response handling, validity, and MAP-local gates are identical for the real signal and every realization. Re-estimation or reselection creates another method.

SCI-NOI-ASM-004 - target-law complement symmetry

ODQ-102C supplies a complement-symmetric network-stratified coefficient-mass balance family with marginal detector sign probability 1/2. It does not supply the detailed finite design, realized independence, count balance, or pixelwise source cancellation.

SCI-NOI-ASM-005 - detailed design remains pending

ODQ-102D delegates an implementation-blind author proposal but does not accept its mechanics in advance. Tolerance, population edge cases, candidate law, search, retry, replacement, complement occurrence, duplicate/equivalence, rank, weights, count domain, generator/keying, and observation-product scope remain unavailable until the owner accepts one exact complete candidate.

SCI-NOI-ASM-006 - conditional target only

The fixed parent can contain astronomical source and structured residuals. The conditional detector-sign law is not assumed equal to repeated physical noise, total MAP uncertainty, a source-free null, or a calibrated tail law.

SCI-NOI-ASM-007 - atomic completion

ODQ-105A supplies the all-admitted-member completion rule. Candidate construction rejection is not member failure. Once admitted, member failure fails the ensemble for base UNC; diagnostics do not become survivor inference.

SCI-NOI-ASM-008 - no distributional significance model

No Gaussian, Student, pivotal, random-field, search, selection, multiplicity, false-alarm, completeness, purity, or catalog model is supplied.

SCI-NOI-ASM-009 - immutable successors

Adjacent parents and prior NOI products are immutable. Owner-defined transformation, Wiener, and FRUIT routes require exact external authority and new generations whenever learning, selection, continuation, or replay changes.

SCI-NOI-ASM-010 - profile revision boundary

The four exact Stage A NOI profile bytes are registered. r0.2 symbol or action changes do not silently rewrite them. Any affected r0.2 evaluation remains unavailable until an exact immutable profile/Registry successor is approved and bound.

SCI-NOI v0.1 r0.2 Shared Normative Module: Requirements

Module identity: SCI-NOI_NORMATIVE_REQUIREMENTS v0.1/r0.2

Status: implementation-blind Stage B draft authority; content-bound by
SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2; not owner-accepted or frozen.

Normative terms: shall is required, shall not is prohibited, may is permitted only within the stated boundary, and unavailable is a typed scientific state rather than a numerical sentinel.

SCI-NOI-REQ-001 NOI-GEN, NOI-UNC, and NOI-STD shall remain separate scientific operations. No operation shall automatically realize, authorize, or mutate another.

SCI-NOI-REQ-002 The selected ordinary route shall remain
NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1, with exact retained PTC occurrences as earliest immutable parent and the exact PTC-to-MAP boundary as insertion point. Its output shall be an NOI realization map, not ordinary MAP science.

SCI-NOI-REQ-003 Every GEN method shall classify consequential adjacent state as fixed, rerun/relearned, not applicable, or unavailable. Different fixed/ relearned classes or replay graphs shall not share an ensemble.

SCI-NOI-REQ-004 The ordinary coherence unit shall be one stable realized detector/channel in one exact observation. Its modifier shall apply to every admitted occurrence for that detector in that observation and shall be ordered by canonical scientific identity rather than execution order.

SCI-NOI-REQ-005 Network-stratified balance shall use exact frozen positive pre-normalization coefficient mass β_d from SCI-NOI-EQ-003. No network, array, or observation shall balance another, and β_d shall not be promoted to precision, uncertainty weight, exposure, validity, or a MAP coefficient.

SCI-NOI-REQ-006 Base product scope shall remain unavailable pending SCI-NOI-OWNER-Q-SCOPE-01. The r0.2 candidate is one observation-level ensemble per exact observation and array, with no cross-observation ensemble or randomized coadd. A multi-observation/coadd GEN route shall require a separate method, parent, assignment law, response, coefficient, and atomicity contract.

SCI-NOI-REQ-007 Every ordinary member shall satisfy SCI-NOI-EQ-005. The modifier shall act only on Z_i^{PTC} , exactly once. Every coefficient, membership, denominator, WCS, projection, support, coverage_cut, response rule, and MAP-local gate shall remain frozen.

SCI-NOI-REQ-008 Inline and materialized sign application shall be equivalent only when they realize the identical equation, occurrence membership, modifier, frozen operator, and output domain. The all-+1 operation shall retain the distinct ordinary MAP identity in SCI-NOI-EQ-006.

SCI-NOI-REQ-009 The fixed-state signal/response relation shall satisfy SCI-NOI-EQ-007. A required response companion shall receive identical membership and the identical modifier exactly once. Missing response shall be typed unavailable and shall not create a hidden signal subset.

SCI-NOI-REQ-010 Products shall distinguish the complete coherence population, positive-mass active population, zero-mass detectors, zero-total- mass strata, singleton active strata, and unavailable/ambiguous network membership. No imbalance ratio shall be evaluated with a zero denominator.

SCI-NOI-REQ-011 Zero-mass, zero-total, singleton, ambiguous-membership, and active-sign-matrix treatment shall remain unavailable pending acceptance of the complete finite-design disposition. The r0.1 singleton failure prediction shall not be retained as binding owner-approved science.

SCI-NOI-REQ-012 Exact ODQ-102D finite-design mechanics shall remain unavailable until the owner answers SCI-NOI-OWNER-Q-102D-01 for the complete candidate as one indivisible disposition. Reviewer recommendations and author proposals shall not become base-v0.1 defaults.

SCI-NOI-REQ-013 Before candidate or member output is inspected, an accepted design plan shall preregister every tolerance, requested count, retry cap, generator/version, seed/key bytes, namespace, persistence mode, ordering, and target-law identity. Any post-inspection change shall create a new plan and generation.

SCI-NOI-REQ-014 If the recommended bounded first-accepted rejection design is owner-approved, finite retry shall affect design-resolution probability but, conditional on successful resolution, shall not change the uniform law over the admissible sign vectors. This conditional result shall not imply approval of the design.

SCI-NOI-REQ-015 Complement-related assignments shall remain distinct; no complement pair shall be forced to occur. Exact duplicates, complement orbits, replacement rules, and draw weights shall remain separately declared under an accepted design.

SCI-NOI-REQ-016 Products shall report requested, resolved, completed, admitted, signed-vector-unique, and complement-orbit-unique counts separately. They shall report `r_sign`, `r_map`, and use-specific effective Monte Carlo information separately; none shall substitute for another.

SCI-NOI-REQ-017 The product class source-bearing conditional randomization ensemble shall carry the narrow objective and exclusions in D-005. Complete complement symmetry shall establish zero target-law mean only; it shall not remove source or structured residual terms from members, second moments, or tails.

SCI-NOI-REQ-018 Every admitted assignment shall complete through the exact declared frozen operator. Any incomplete, failed, or unavailable admitted member shall fail the ensemble for initial UNC. Survivors and partial accumulators shall have diagnostic identity only.

SCI-NOI-REQ-019 An idempotent execution retry may retain scientific member, assignment, parent, operator, and generation identity only when deterministic work is identical and a new execution/evidence attempt identity is recorded. A changed assignment or scientific state shall be a replacement with new scientific identity.

SCI-NOI-REQ-020 Each ODQ-109 persistence plan shall select persisted, compact-regeneration, streaming-sufficient-statistic, or an explicitly separate composite plan. It shall declare exact reconstruction, numerical regeneration, retained sufficiency, unavailable later questions, requested/ effective/resolved/applied/realized lifecycle, failure, and completion. No default or silent fallback is permitted.

SCI-NOI-REQ-021 An externally owned deterministic transform shall be used only under exact owner authority and parity for purpose, algorithm, state, parameters, order, domain, support/edge/missing-data behavior, normalization, units, response, validity, lifecycle, and failure. NOI shall not choose, tune, substitute, relocate, simplify, or silently omit it.

SCI-NOI-REQ-022 Wiener or FRUIT learning, selection, continuation, update, or per-member replay shall create every applicable new immutable owner, science-product, GEN, and UNC generation. Fixed and relearned members shall not mix. All base-v0.1 numerical transformed routes remain unavailable pending their exact owner boundaries.

SCI-NOI-REQ-023 Every UNC method shall bind exact target law, admitted GEN generation/membership, center, estimator, weights, dependence/missingness, domain, WCS/support/response/unit, representation, both applicable ranks, regularization, calibration/omissions, estimator uncertainty, lifecycle, claim, and named use.

SCI-NOI-REQ-024 Initial UNC shall consume only an exact all-members- successful ensemble admitted by the named member and ensemble profiles. Rejected candidates, failed ensembles, survivors, pairwise subsets, mixed methods, failed persistence, and partial streaming state shall not be admitted.

SCI-NOI-REQ-025 Initial UNC shall compute SCI-NOI-EQ-009 on one exact common all-member domain. Its target shall be the conditional detector-sign law with zero target mean from complement symmetry. It shall not subtract the finite ensemble mean or infer a B or B-1 correction.

SCI-NOI-REQ-026 `Vhat_cond` shall have squared signal units and shall be labeled conditional detector-sign-randomization marginal variance/second moment for the fixed declared target only. It shall not be repeated physical- noise variance, total MAP uncertainty, covariance completeness, precision, or significance. Source and deterministic residual imprint shall remain explicit.

SCI-NOI-REQ-027 Estimator uncertainty for `Vhat_cond` shall be numerical and explicit or typed unavailable. Member count, sign uniqueness, complement-orbit uniqueness, `r_sign`, `r_map`, and effective Monte Carlo information shall not be conflated with that uncertainty.

SCI-NOI-REQ-028 Retained ensemble, marginal variance, fixed projection, structured/full covariance, reciprocal, precision, consumer weight, and unavailable representations shall remain separately versioned roles. Missing covariance shall be unknown/unavailable rather than zero or independence.

SCI-NOI-REQ-029 The r0.2 reciprocal successor shall remain unavailable pending SCI-NOI-OWNER-Q-107-NAME-01 and an immutable profile/source successor. If accepted, it shall use SCI-NOI-EQ-010 only on the exact finite strictly positive parent domain and shall not be inverse variance, precision, validity, support, exposure, a consumer weight, or a PTC/MAP coefficient.

SCI-NOI-REQ-030 The initial STD method shall use `m_MAP`, `sigma_cond`, and `zeta_cond` exactly as SCI-NOI-EQ-011 and SCI-NOI-EQ-012 on the compatible finite-positive intersection. It shall not interpolate, extend, substitute, regularize, or create an implicit reciprocal route. JINC remains separate and unavailable.

SCI-NOI-REQ-031 `zeta_cond` shall have unit exactly 1, explicit numerator/ scale dependence, and the sole claim: "MAP signal standardized by the stated conditional detector-sign-randomization second-moment scale." No Gaussian, Student, pivotal, `z`, `N-sigma`, probability, detection, false-alarm, completeness, purity, or catalog claim shall follow.

SCI-NOI-REQ-032 Every NOI admission evaluation shall retain separately named request, applicability, eligibility, and realization fields. Only the exact positive conjunction shall project to one exact action. A generic flag or another-use result shall have no universal veto, rescue, or propagation.

SCI-NOI-REQ-033 Generation-input admission shall permit only one exact candidate occurrence for the selected GEN method. Assignment, MAP application, completion, ensemble completion, and UNC admission shall remain separate.

SCI-NOI-REQ-034 Uncertainty-member admission shall consume but not redefine GEN completion, failure, duplicate/equivalence, support, source-imprint, QC, persistence, lifecycle, cause, and provenance facts. It shall admit only one candidate member for one named UNC method.

SCI-NOI-REQ-035 Uncertainty-ensemble admission shall permit only one exact complete all-members-successful ensemble for one exact estimator/domain. It shall not realize the estimator or create another covariance/consumer use.

SCI-NOI-REQ-036 Standardization admission shall permit only the exact initial MAP numerator/scale join and claim. Because r0.2 notation changes the registered Stage A action bytes, r0.2 evaluation shall remain unavailable until an immutable profile/Registry successor is approved and bound.

SCI-NOI-REQ-037 Missing, conflicting, incomplete, unsupported, invalid, or failed required facts shall yield their typed state. No undocumented zero, NaN, infinity, empty object, filename, equal shape, approximate WCS, or hidden default shall establish success.

SCI-NOI-REQ-038 PTC, MAP, JINC, transform, Wiener, FRUIT, and prior NOI parents shall remain immutable. Every later result shall be a new versioned companion with exact parent, method, generation, dependence, and lifecycle.

SCI-NOI-REQ-039 No NOI assignment, uncertainty, reciprocal, precision, scale, or weight shall become validity, support, exposure, or a PTC/MAP analysis, gridding, or coadd coefficient without separately approved successor authority.

SCI-NOI-REQ-040 The four owner-approved Stage A profiles and exact SCI-VAL successors shall remain immutable. r0.2 shall report their exact bindings and shall not infer compatibility for changed r0.2 bytes.

SCI-NOI-REQ-041 The rationale and ECS shall import or content-bind the exact six shared normative modules. They shall not independently author equations, requirements, or predictions. Any module byte change shall require a new binding digest and document generation.

SCI-NOI-REQ-042 This draft shall establish no implementation conformity, numerical validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization. Every unresolved route shall remain unavailable.

SCI-NOI v0.1 r0.2 Shared Normative Module: Predictions

Module identity: SCI-NOI_NORMATIVE_PREDICTIONS v0.1/r0.2

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.2; not owner-accepted or frozen. These are analytic expectations of the stated contract, not reports of implementation conformity or empirical validation.

SCI-NOI-PRED-001 A globally complemented assignment has equal probability under the approved complement-symmetric target-law family.

SCI-NOI-PRED-002 Every active detector has target-law marginal sign mean zero; under the fixed linear member equation, every available map row has target-law member mean zero.

SCI-NOI-PRED-003 Complement symmetry does not remove source or structured residual contributions from individual members, conditional second moments, or tails.

SCI-NOI-PRED-004 Under the exact fixed member equation, globally complementing every active sign negates M_b while leaving Q_p unchanged.

SCI-NOI-PRED-005 Complement-related members have identical pointwise squares and outer products even though their full-distribution identities remain distinct.

SCI-NOI-PRED-006 $V_{\text{hat_cond}}$ is nonnegative wherever available and is unavailable outside the exact common all-member domain.

SCI-NOI-PRED-007 A source-bearing fixed component can create source-shaped structure in $V_{\text{hat_cond}}$; zero target-law mean is insufficient to prevent it.

SCI-NOI-PRED-008 Any failed admitted member makes the initial UNC ensemble unavailable; completed survivors do not realize a partial estimator.

SCI-NOI-PRED-009 Increasing member count may improve estimation of the declared randomization target but cannot create astronomical exposure or reduce the immutable parent map's realized noise as $1/\sqrt{N}$.

SCI-NOI-PRED-010 A fixed owner-defined linear transformation propagates a conditional second moment by the fixed operator relation; a per-member learned transformation does not satisfy that identity by implication.

SCI-NOI-PRED-011 A positive decision under one NOI profile neither realizes the next operation nor changes producer-owned facts.

SCI-NOI-PRED-012 If a required response companion is unavailable, response- dependent output is unavailable; signal membership does not silently shrink.

SCI-NOI-PRED-013 An idempotent deterministic execution retry can change attempt/evidence identity without changing scientific member identity; a changed assignment or scientific state cannot.

SCI-NOI-PRED-014 Conditional on later owner acceptance of the recommended independent symmetric candidate law and first-accepted bounded rejection, finite retry changes resolution probability but, conditional on success, not the uniform distribution over admissible sign vectors.

SCI-NOI-PRED-015 A zero-total-mass stratum cannot support an imbalance ratio; its scientific treatment remains unavailable pending owner disposition rather than producing a numerical zero-denominator result.

SCI-NOI-PRED-016 No binding singleton-active-stratum feasibility or failure prediction exists until the owner accepts its exact design treatment.

SCI-NOI-PRED-017 zeta_cond has unit exactly 1 on its available domain; dimensionlessness does not establish a pivotal, Gaussian, Student, z , N -sigma, probability, detection, false-alarm, completeness, purity, or catalog interpretation.